WORKTYPE

AI/ML
Multi-modal Learning
Medical Imaging

PORTFOLIO

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EDUCATION

2024.03 - Present POSTECH
GPA 3.93/4.30

2019.03 - 2024.02 Inha Unversity
GPA 4.22/4.50

2022.09 - 2023.02 University of Hull, England

RESEARCH INTEREST

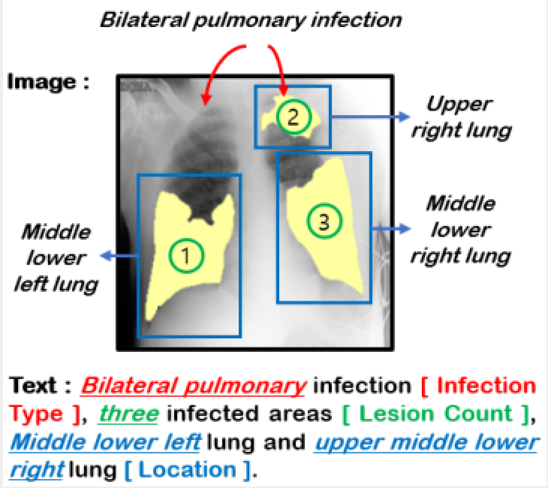
- Multi-modal Learning
- Computer Vision
- Medical Imaging
- Large Language Model

Publication

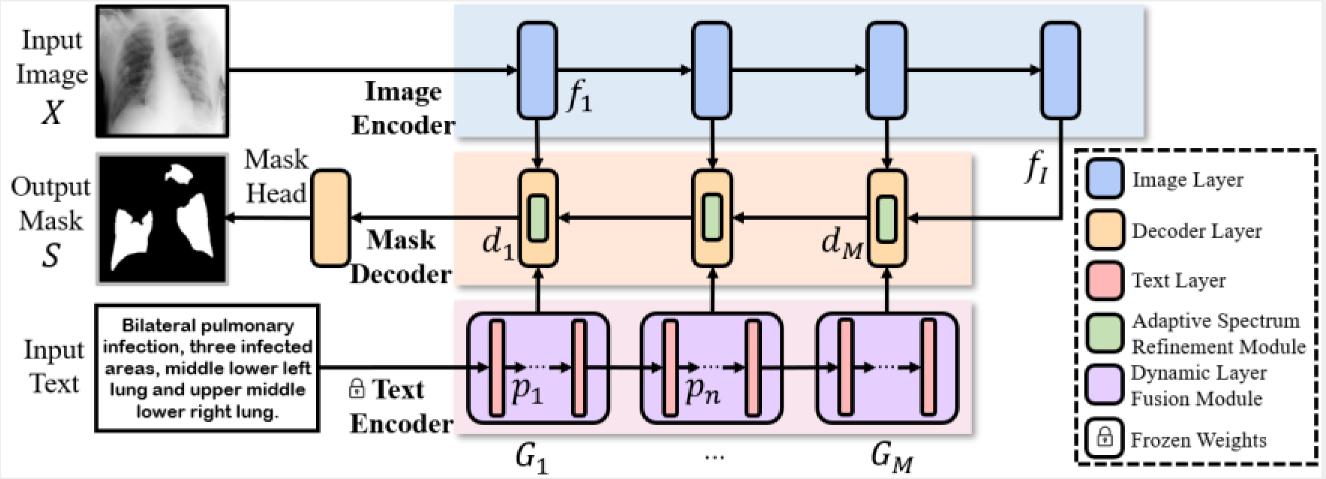
HiMix : Hierarchical Visual-Textual Mixing for Segmentation (WACV 2026)

Soojin Hwang*, Jaeyoon Sim*, Won Hwa Kim

- Overview
  - Most language-guided lesion segmentation methods rely on **a single text embedding**, limiting the use of rich linguistic information.
  - We propose **HiMix**, a **hierarchical visual–textual fusion framework** that aligns multi-level text features with the image decoding process.
- Key contributions
  - Hierarchical Fusion
    - HiMix integrates **multi-level text representations** with **multi-scale visual features** across decoder stages for fine-grained segmentation.
  - Dynamic Layer Fusion (DLFM)
    - Dynamically fuses **multi-layer text embeddings** and aligns them with **corresponding decoder stages.**
  - Adaptive Spectrum Refinement (ASRM)
    - Adaptively refines decoder features in the **frequency domain** to capture **both global context and fine-grained boundaries.**



Example of an image-prompt describing structured radiologic findings



Architecture of HiMix

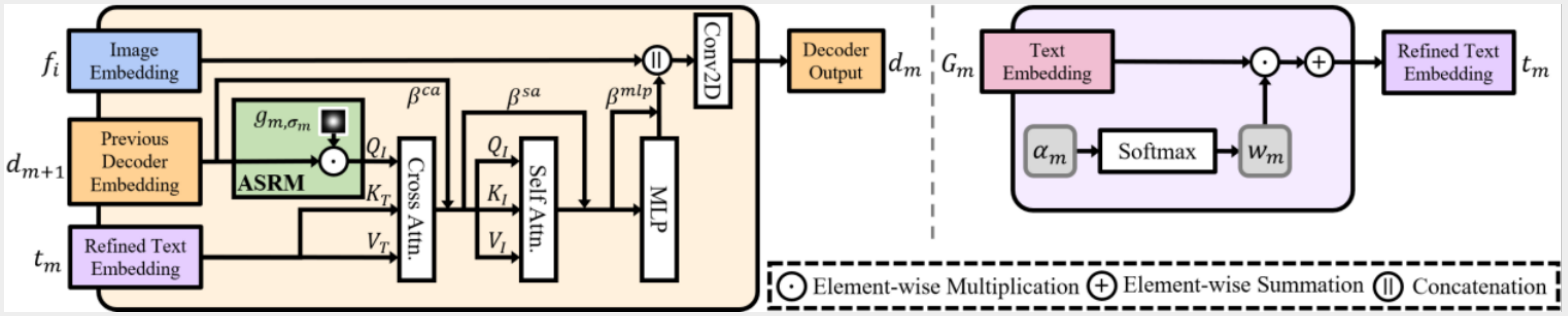


Illustration of key modules in HiMix. Left : Decoder with adaptive spectrum refinement module (ASRM). Right : Dynamic layer fusion module (DLFM)

Publication

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- Quantitative Results
  - Superior Performance: HiMix outperforms both uni-modal and multi-modal baselines, including SOTA models.
  - Few Parameters: HiMix performs well with only 44.7M parameters, significantly fewer than other models.

Approach	Type	Method	Param ↓ (M)	QaTa-COV19		MosMedData+		Kvasir-SEG	
				DSC ↑	IoU ↑	DSC ↑	IoU ↑	DSC ↑	IoU ↑
Uni-Modal (Image-Only)	CNN	UNet [37]	<u>14.8</u>	79.02	69.46	64.60	50.73	82.94	74.47
		UNet++ [56]	74.5	79.62	70.25	71.75	58.39	80.43	72.13
		AttnUNet [33]	34.9	79.31	70.04	66.34	52.82	81.31	73.74
		nnUNet [19]	19.1	80.42	70.81	72.59	60.36	85.06	74.01
	Transformer	Swin-UNet [5]	82.3	78.07	68.34	63.29	50.19	75.97	67.45
	Hybrid	UCTransNet [46]	65.6	79.15	69.60	65.90	52.69	78.21	65.25
		TransUNet [6]	105.0	78.63	69.13	71.24	58.44	79.67	71.14
	Mamba	VMUNet [38]	31.0	86.31	75.92	75.85	61.09	79.54	66.04
		H-vmunet [49]	31.0	86.26	75.84	76.37	61.78	82.25	69.85
Multi-Modal (Image-Text)	CNN	GLoRIA [17]	45.6	79.94	70.68	72.42	60.18	84.73	73.51
		ConVIRT [53]	35.2	79.72	70.58	72.06	59.73	84.27	72.82
		RecLMIS [18]	23.7	85.22	77.00	77.48	<u>65.07</u>	87.73	78.15
	Transformer	CLIP [35]	87.0	79.81	70.66	71.97	59.64	81.27	68.45
		MedCLIP [47]	137.0	86.54	76.27	69.14	52.84	78.84	65.08
		DMMI [15]	131.5	79.25	68.87	72.57	60.78	74.79	59.73
		RefSegformer [48]	195.0	84.09	75.48	74.98	61.70	79.78	66.37
		MedSAM [30]	**4.5**	78.49	69.11	54.22	42.22	86.69	79.24
	Hybrid	ViLT [23]	87.4	79.63	70.12	72.36	60.15	70.33	54.23
		LAVT [51]	118.6	79.28	69.89	73.29	60.41	70.59	54.55
		LViT [27]	29.7	83.66	75.11	74.57	61.33	87.17	77.26
		SLViT [34]	114.6	84.13	75.66	75.01	61.83	86.61	76.38
		GuideDecoder [55]	44.0	89.78	81.45	77.75	63.60	88.31	79.07
		MMI-UNet [4]	56.2	<u>90.88</u>	<u>83.28</u>	<u>78.42</u>	64.50	<u>91.43</u>	<u>84.57</u>
		HiMix (Ours)	44.7	**91.17**	**83.78**	**79.44**	**65.90**	**92.18**	**85.50**

The best and second-best results are highlighted in **bold** and <u>underlined</u>, respectively

Publication

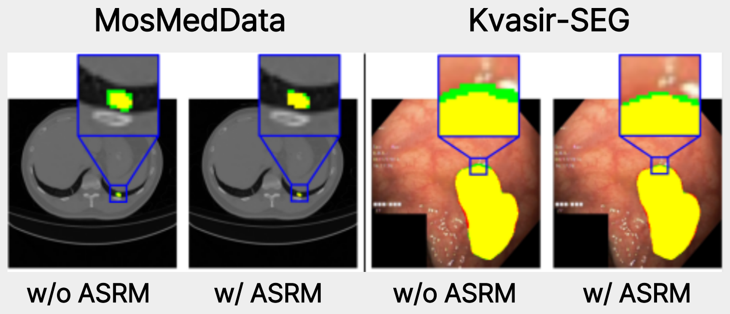
HiMix : Hierarchical Visual-Textual Mixing for Segmentation (WACV 2026)

Soojin Hwang*, Jaeyoon Sim*, Won Hwa Kim

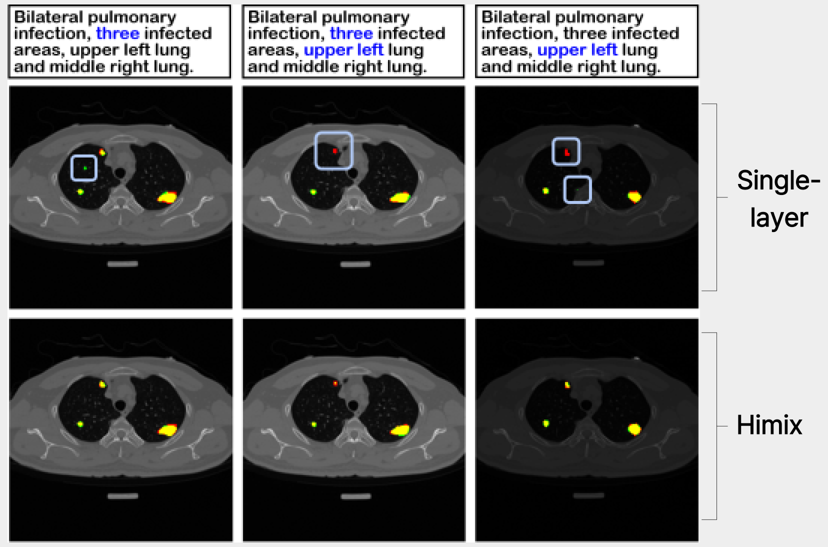
- Key Findings
  - **Hierarchical Alignment:** Aligning hierarchical text features with decoding stages enables more accurate and context-aware segmentation.
- Conclusion
  - HiMix shows that **hierarchical text-image fusion improves the fine-grained lesion segmentation** across diverse medical datasets .

Additional Analysis

(1) ASRM Ablation



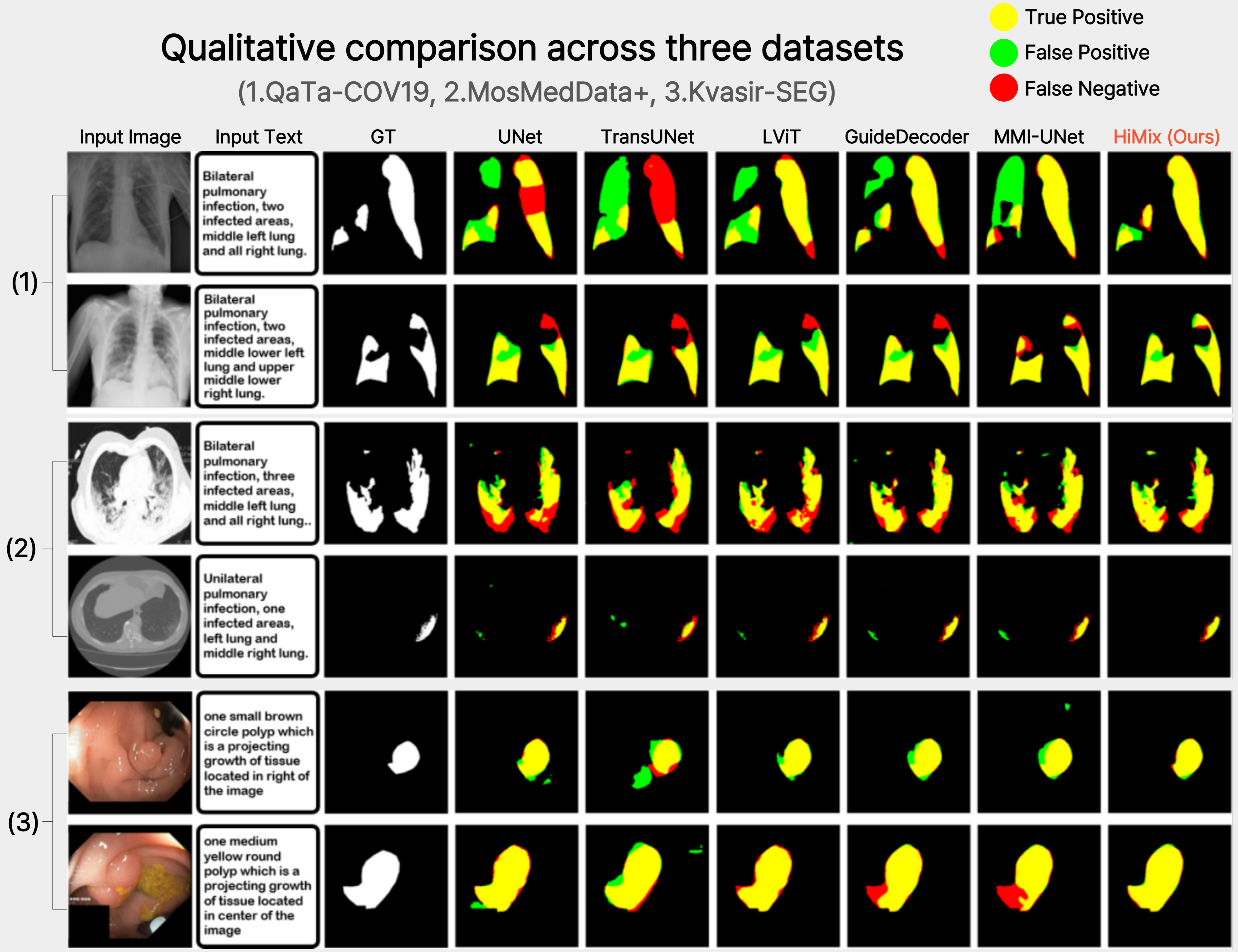
(2) Hierarchical Text Fusion



Both (1) Spectrum refinement and (2) Hierarchical text fusion improve segmentation accuracy and reduce over-segmentation.

Qualitative comparison across three datasets

(1.QaTa-COV19, 2.MosMedData+, 3.Kvasir-SEG)



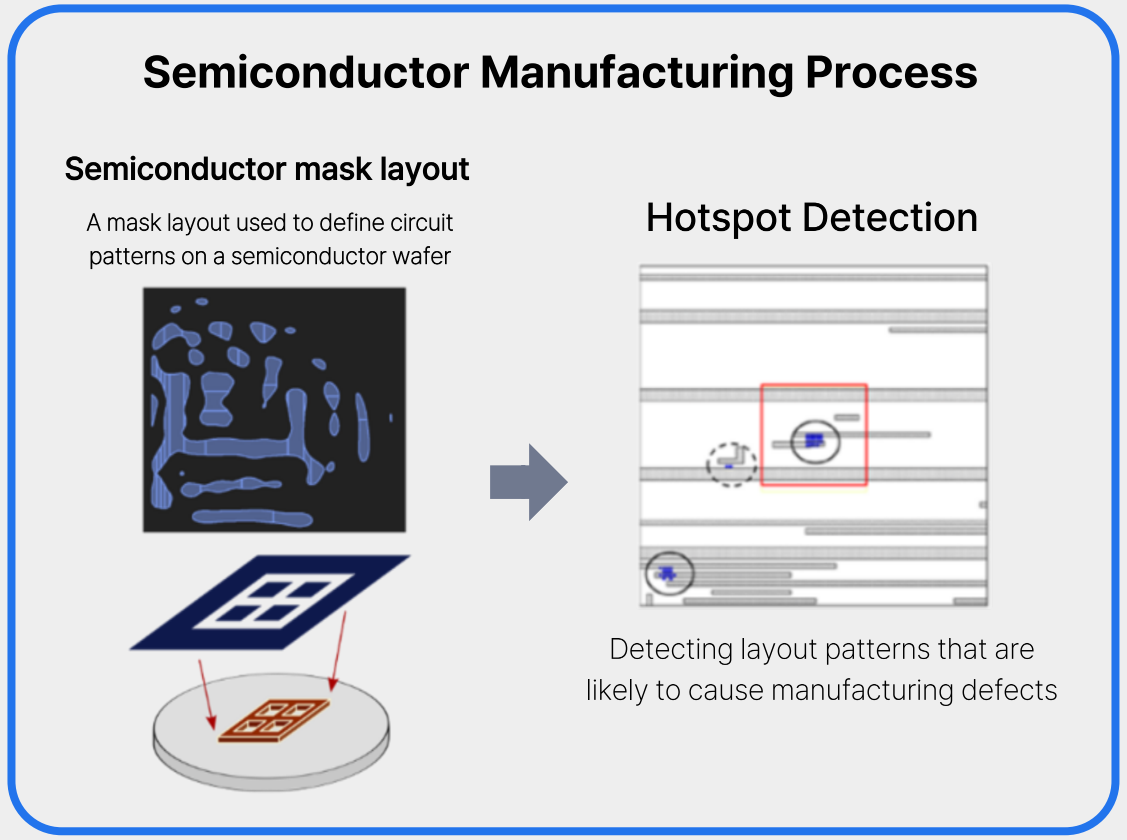
HiMix produces more accurate and consistent lesion segmentation than prior language-guided models.

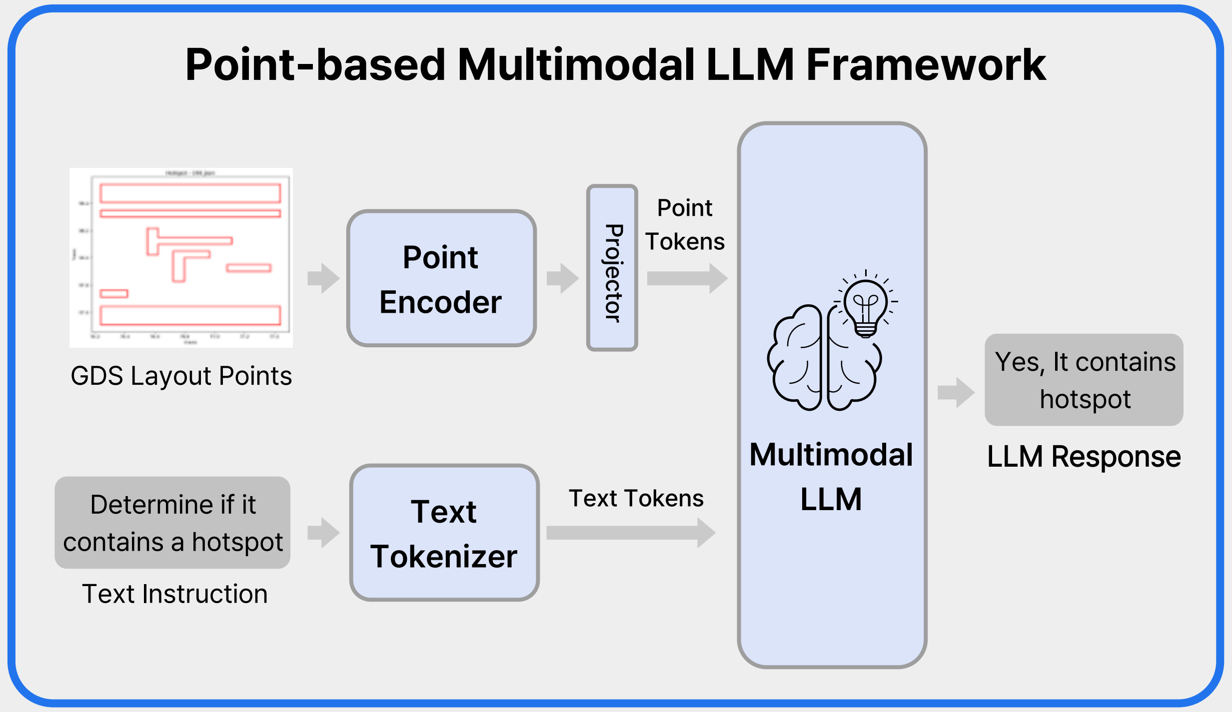
Project

Point-based AI Framework for Semiconductor Layout Analysis

Joint Research with Samsung Electronics DS AI center (2025.04 - 2026-02)

- Overview
  - Existing approaches for **semiconductor layout hotspot detection** typically **rasterize GDS (Graphic Design System) layouts into images**, which may distort fine geometric structures.
  - We explored two approaches : (1) **text-based** and (2) **point-based multimodal approach.**
- Key Contributions
  - (1) Text-based LLM Reasoning
    - Converted **GDS layouts into structured text prompts**
    - Constructed **instruction-sample datasets** for hotspot detection
    - Proposed **Mixture-of-Prompts (MoP)** for adaptive prompt selection
  - (2) Direct Point-based Multimodal Learning
    - Adapted **Point-BERT** to learn geometric features from GDS layout polygon points
    - Designed a **point-based representation** for GDS layouts
- Impact
  - Built a **scalable Pytorch training pipeline** for large-scale experiments (GPU A6000, A100, H100)
  - Demonstrated the **feasiblity of point-based geometric learning** for semiconductor layout task
  - Explored **geometry-based** and **language-based** approaches for layout hotspot detection





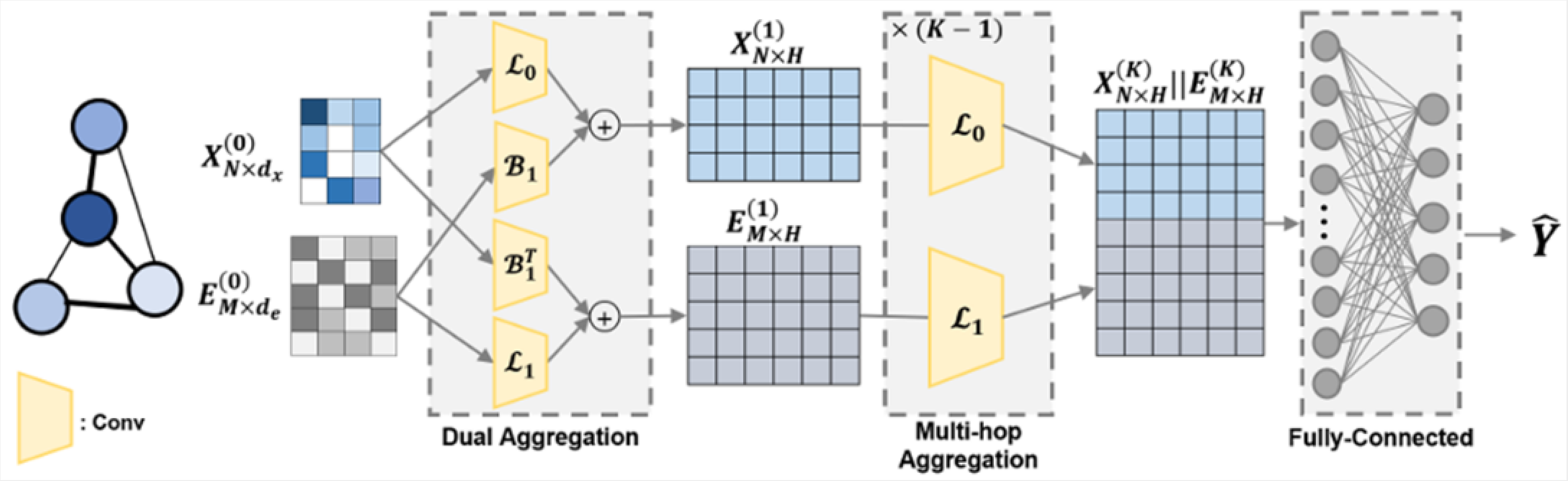
Publication

Multi-order Simplex-based GNN for Brain Networks (MICCAI 2024)

Yechan Hwang, Soojin Hwang, Guorong Wu, Won Hwa Kim

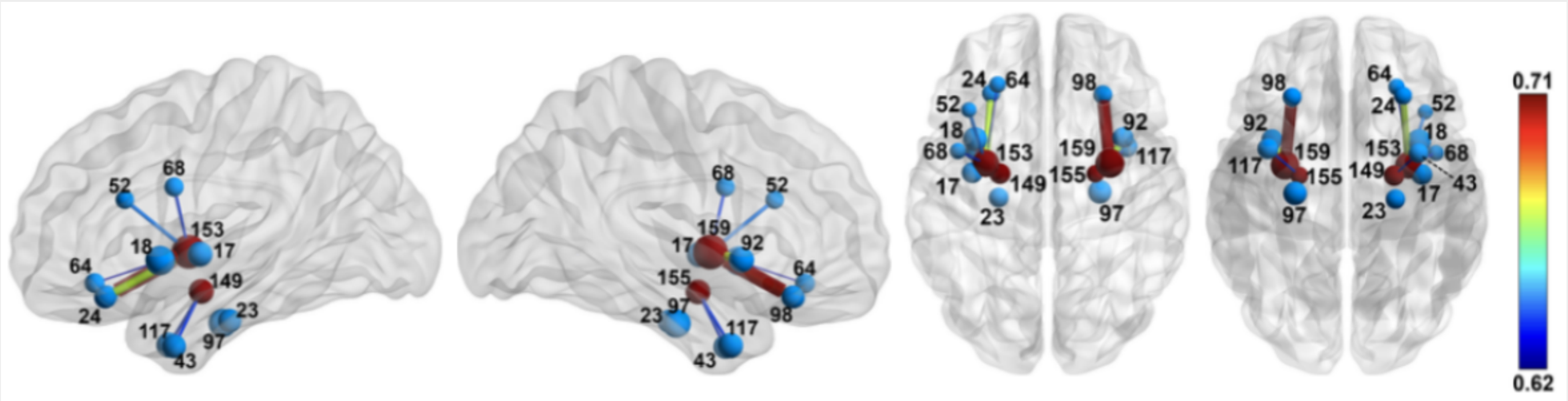
- Overview
  - Existing GNN-based brain network models focus on node-centric analysis, treating edge features as auxiliary information.
  - We propose a **topology-aware Graph Neural Network** that jointly learns node and edge representations for **Alzheimer's disease (AD) classification.**
- Key Contributions
  - **Dual aggregation framework for node-edge interactions in brain networks.**
  - Topology-aware graph learning using **Incidence matrix and Hodge Laplacian.**
  - Interpretable analysis of brain regions and connectomes associated with AD
- Results
  - Achieved **state-of-the-art performance** on the Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, reaching up to 93.5% accuracy and outperforming existing GNN models (GCN, EGNN, CensNet).

Overview of the proposed topology-aware GNN architecture



Grad-CAM visualization highlighting key brain regions for AD classification

(Node size and edge color indicate activation strength.)



Left hemisphere Right hemisphere Top View Bottom View

The model highlights AD-related brain regions such as the putamen, amygdala, and parahippocampal gyrus.

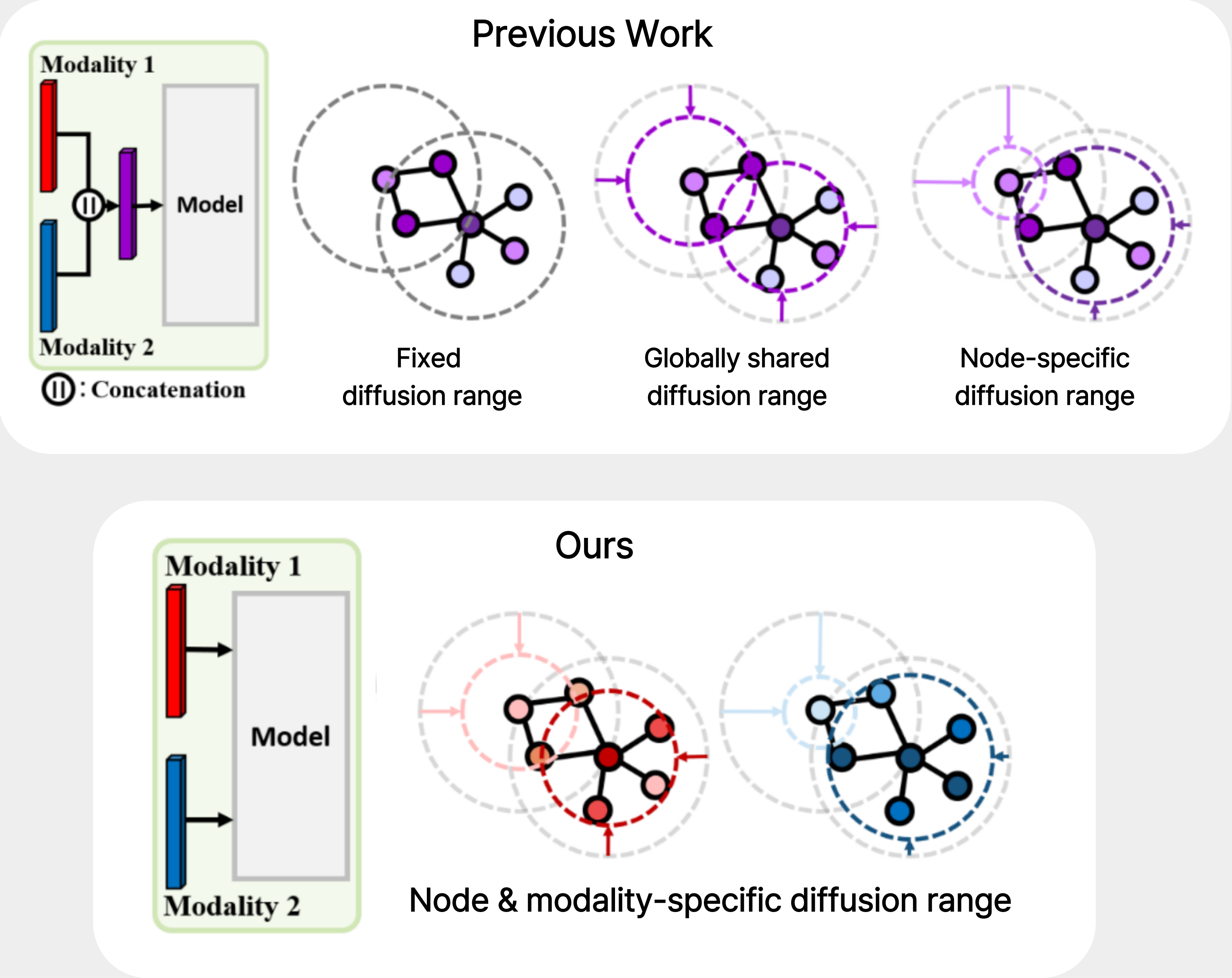
Publication

GTAD : Transformer-Guided Adaptive Diffusion for Multi-Modal Brain Networks (under review)

Jaeyoon Sim*, Soojin Hwang*, Guorong Wu, Won Hwa Kim

- Overview
  - Most brain network GNNs struggle to capture both **local patterns** and **global dependencies** across multi-modal neuroimaging data (e.g., MRI and PET biomarkers).
  - We propose **GTAD**, a transformer-guided diffusion graph network that learns **modality-aware brain representations** for Alzheimer's Disease (AD) classification.
- Key Contributions
  - Adaptive Graph Diffusion
    - Learns **node- and modality-specific diffusion scales** using heat kernel graph diffusion to capture **adaptive local connectivity patterns.**
  - Multi-Modal Transformer Attention
    - Applies self-attention to model **long-range dependencies** between brain regions across modalities.
  - Modality-Aware Brain Modeling
    - Processes each neuroimaging modality separately to preserve **modality-specific biomarkers and interpretability.**

Previous Work vs. Ours



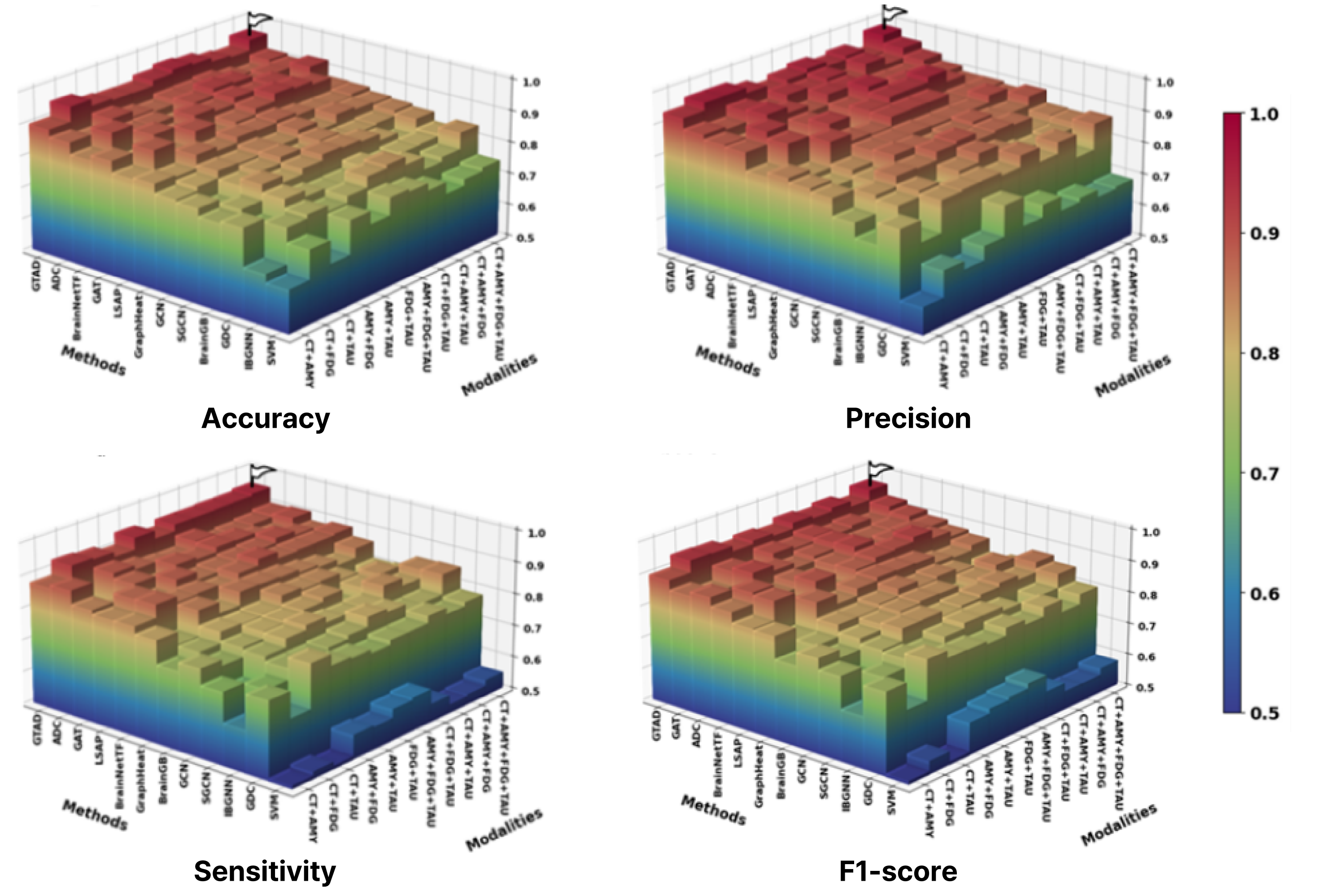
Publication

GTAD : Transformer-Guided Adaptive Diffusion for Multi-Modal Brain Networks (under review)

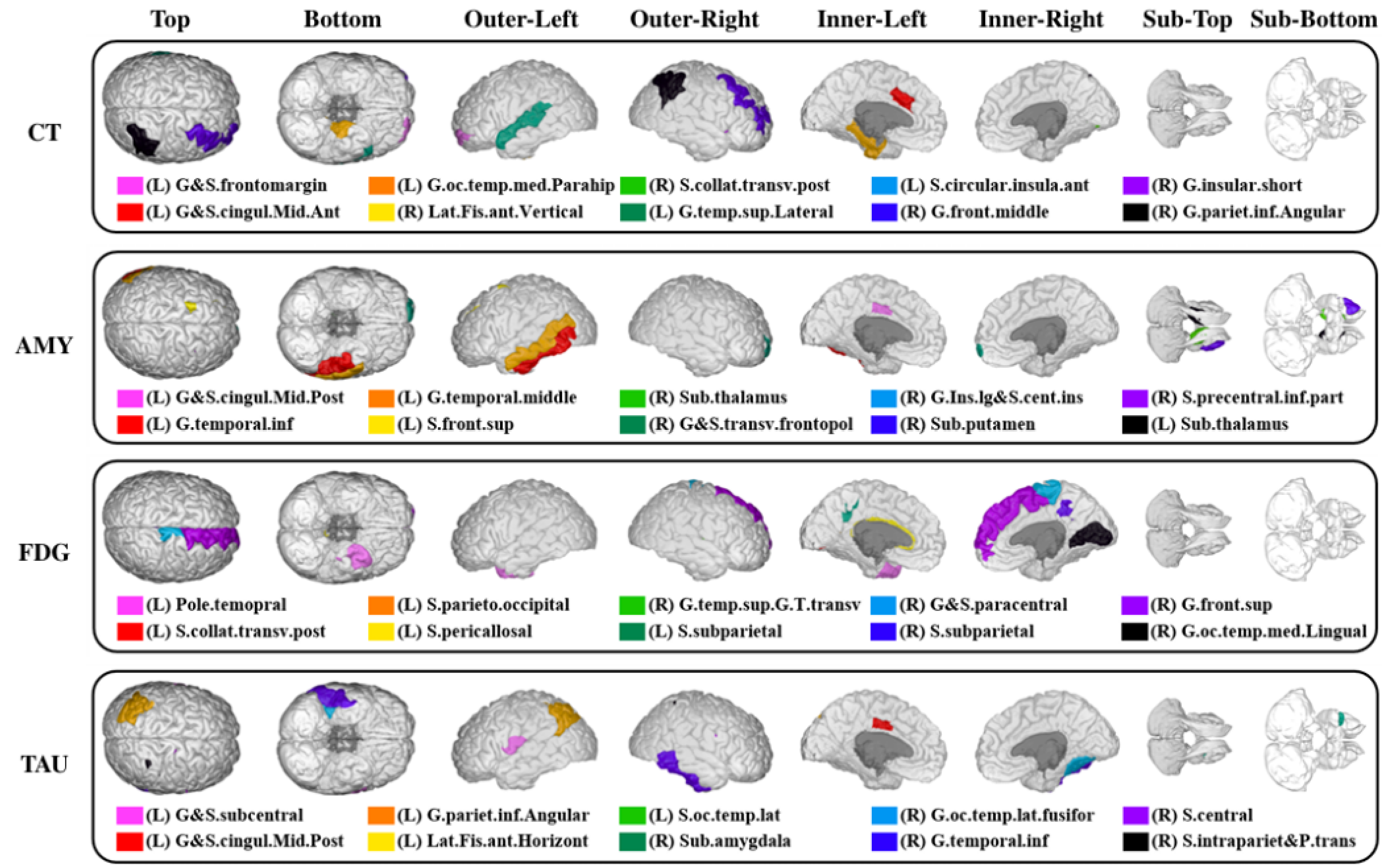
Jaeyoon Sim*, Soojin Hwang*, Guorong Wu, Won Hwa Kim

- Results
  - GTAD outperforms state-of-the-art GNN models across multi-modal brain networks.
  - Best performance with all modalities (Acc 0.948, Prec 0.065, Sens 0.939, F1 0.948).
  - Provides interpretable insights by identifying key ROIs related to AD.

Performance comparison across models and modality combinations



Visualization of ROIs with the 10 highest attention scores for each modality



Project

Real-Time PPE Monitoring System using AWS Video Pipeline

Winter Internship, KIST(Korea Institute of Science and Technology) (2021.12 - 2022-02)

- Overview
  - Built a **cloud based AI video analysis pipeline** using AWS services (Kinesis Video Stream, Amazon Rekognition, S3)
  - Enabled real-time video streaming from a Double 3 robot camera using GStreamer on Ubuntu.
- Key Contributions
  - Implemented **PPE(Personal Protective Equipment) detection and face recognition** using Amazon Rekognition
  - Built a **real-time video streaming pipeline** using **GStreamer and Kinesis Video Streams**
  - Integrated AWS services (S3, Rekognition, Kinesis) to automated video ingestion and AI-based analysis
- Impact
  - Developed a **real-time PPE monitoring system** capable of detecting safety equipment compliance from robot-captured video streams.

PPE (Personal Protective Equipment)

double 3 robot

Thank you for your time

AI Researcher/Engineer Porfolio
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Appendix

GTAD : Transformer-Guided Adaptive Diffusion for Multi-Modal Brain Networks (under review)

Jaeyoon Sim*, Soojin Hwang*, Guorong Wu, Won Hwa Kim

An overview of the proposed GTAD for Alzheimer's diagnosis and biomarker interpretation.